

GOKULAKKANNAN L

AI / MACHINE LEARNING ENGINEER — COMPUTER VISION

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PROFESSIONAL SUMMARY

Aspiring Computer Vision and Machine Learning Engineer with 6 years of professional experience in visual effects (VFX) camera tracking and matchmove, bringing deep expertise in image processing, spatial analysis, and precision visual data interpretation. Recently completed a specialized AI Developer diploma covering Python, Machine Learning, Deep Learning, and Computer Vision. Skilled in building end-to-end computer vision and ML solutions using TensorFlow, Keras, OpenCV, and Scikit-learn. Seeking to apply strong analytical and visual-data experience to a Computer Vision Engineer or Machine Learning Engineer role.

TECHNICAL SKILLS

Programming Languages: Python

Computer Vision: OpenCV, Image Classification, Object Detection, Image Processing

Machine Learning: Scikit-learn, Supervised & Unsupervised Learning, Regression, Classification, Feature Engineering, Model Evaluation

Deep Learning: TensorFlow, Keras, CNN, Transfer Learning

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Plotly

Database: SQL, MySQL

Web & Deployment: Streamlit

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab

PROJECTS

AIRA — Artificial Intelligence Rotomation & Animation Pipeline | VFX × AI (Ongoing)

- Developed a VFX-focused AI pipeline concept for automated 3D rotomation and character motion extraction from video footage.
- Designed a workflow combining computer vision, pose/motion extraction, 3D character rigging and motion retargeting to generate animation from live-action footage.
- Planned multi-output rendering including original footage with the animated 3D character and motion/animation data for further VFX workflows.

Multiclass Fish Image Classification — Deep Learning

- Built and trained a multiclass image classification model using TensorFlow/Keras and transfer learning to classify fish species from images.
- Achieved 99.53% test accuracy on the final model after dataset preparation, preprocessing and model optimization.
- Developed a Streamlit-based interface for image-based prediction and prepared the project for deployment with documented dependencies and usage instructions.

Brain Tumor Image Classification — Deep Learning (Ongoing)

- Developed a multiclass brain MRI image classification pipeline using Python, OpenCV and TensorFlow/Keras.
- Prepared and balanced image datasets across glioma, meningioma, no-tumor and pituitary tumor classes with image preprocessing and resizing for deep learning.
- Implemented transfer-learning-based classification workflow with preprocessing, label encoding and model evaluation for automated tumor-category prediction.

PROFESSIONAL EXPERIENCE

VFX Artist — Camera Tracking / Matchmove

6 Years Professional Experience

- Performed precision camera tracking and 3D matchmove for VFX production, requiring detailed spatial analysis, pattern recognition, and frame-accurate image processing — core skills directly transferable to computer vision.
- Analyzed thousands of video frames for motion, depth, and perspective consistency, building strong intuition for visual data quality used in CV model evaluation.
- Managed complex production pipelines under strict deadlines, collaborating with cross-functional VFX teams and maintaining rigorous quality control standards.
- Currently transitioning this visual-analysis expertise into Computer Vision, AI, and Data Science through applied Python/ML projects.

EDUCATION

Diploma — AI Developer: Cultus Education (TNSkills), Coimbatore — 2025–2026, Score: 70%

B.Sc. Computer Science: Alagappa University, Erode — 2020–2022, Score: 75%

B.A. Visual Effects and Animation: Mahatma Gandhi University — 2013–2016, Score: 80%